

MAM2000W
Inner Product Spaces

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Contents

Chapter 1	Inner Product Spaces	Page 2
1.1	Inner Product Spaces	2
	Properties of Inner Products — 4	
1.2	Orthogonality and Orthonormal Bases	5

Chapter 1

Inner Product Spaces

1.1 Inner Product Spaces

Note:-

Inner Product is a generalisation of the dot product in $\mathbb{R}^2$ and $\mathbb{R}^3$

Properties of Dot Product:

- $\mathbf{x} \cdot \mathbf{y} = (x_1, x_2, \dots, x_n) \cdot (y_1, y_2, \dots, y_n) = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$
- $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$
- $(\mathbf{x} + \mathbf{y}) \cdot \mathbf{z} = \mathbf{x} \cdot \mathbf{z} + \mathbf{y} \cdot \mathbf{z}$
- $(\lambda \mathbf{x}) \cdot \mathbf{y} = \lambda (\mathbf{x} \cdot \mathbf{y})$
- $\mathbf{x} \cdot \mathbf{x} \geq 0$
- $\mathbf{x} \cdot \mathbf{x} = 0 \iff \mathbf{x} = \mathbf{0}$

Note:-

Inner Product or Inner Product Space?

Definition 1.1.1: Inner Product

Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$.

A function which assigns to any ordered pair $\mathbf{u}, \mathbf{v} \in V$ a scalar $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}$ or $\mathbb{C}$ is an inner product on V if the following hold

- $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$
- $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
- $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle$
- $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0 \iff \mathbf{u} = \mathbf{0}$

Note:-

If we deal with a real vector space, then condition 1 is: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$

(Recall that the complex conjugate of a complex number $z = a + ib$ is given by $\bar{z} = a - ib$.) This means that the inner product on a complex inner product space is no longer symmetric. You may be somewhat surprised by this condition and wonder what the motivation is for this. Here is part of the story: If we take $\mathbf{v} - \mathbf{u}$ in the condition 1' above, we get

$$\langle \mathbf{u}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{u} \rangle},$$

from which we can deduce that $\langle \mathbf{u}, \mathbf{u} \rangle \in \mathbb{R}$; if this were not the case, the condition 4 would not hold.

Much of the theory of real and complex inner product spaces is the same as that for real inner product spaces. In what follows we'll point out specifically where there is a difference between the real and complex case. In the examples in the rest of this chapter we'll concentrate mainly on real inner product spaces, but will mention complex spaces occasionally.

Note:-

If there is an inner product defined on a vector space, we say the vector space is equipped with an inner product. Any vector space equipped with an inner product is called an inner product space.

Note:-

$\langle \mathbf{u}, \mathbf{u} \rangle$ is real even if $\mathbf{u}$ is complex which is needed for comparison $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ since we cannot compare with complex numbers

We now have a third operation on vector spaces:

$$\langle \rangle : V \times V \rightarrow \mathbb{R} \text{ or } \mathbb{C}$$

Example 1.1.1

$V = \mathbb{R}^n$, then $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y}$

Example 1.1.2

$V = C[0, 1]$ (continuous $\mathbb{C}$ shows this) real-valued functions on $[0, 1]$

Then define $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$ is an inner product

We have that

- $\langle f, g \rangle = \langle g, f \rangle$
- $\langle f + g, h \rangle = \int_0^1 (f + g)(x)h(x) dx$
 $= \int_0^1 f(x)h(x) dx + \int_0^1 g(x)h(x) dx = \langle f, h \rangle + \langle g, h \rangle$

Example 1.1.3

$V = M_{m \times n}(\mathbb{R})$. Defining $\langle A, B \rangle = \text{tr}(A^T \cdot B)$

Prove this is an inner product

1.1.1 Properties of Inner Products

Theorem 1.1.1

1. $\langle \mathbf{0}, \mathbf{u} \rangle = \langle \mathbf{u}, \mathbf{0} \rangle = 0$

Note:-

This is the zero in vector space

2. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$

3. $\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \bar{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle$

4. $\langle \mathbf{u} - \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle$

5. $\langle \mathbf{u}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle$

6. If $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all $\mathbf{w} \in V$, then $\mathbf{v} = \mathbf{0}$

Proof: 1. $\langle \mathbf{0}, \mathbf{u} \rangle = \langle 0 \cdot \mathbf{u}, \mathbf{u} \rangle = 0 \langle \mathbf{u}, \mathbf{u} \rangle = 0$

2.

3. $\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \bar{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle$

Note:-

That is λ conjugate



Definition 1.1.2

Let V be an inner product space and let $\mathbf{v} \in V$. The norm of $\mathbf{v}$ (denoted by $\|\mathbf{v}\|$) is given by $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$

Note:-

In $\mathbb{R}^2$ and $\mathbb{R}^3$, the norm is the length of the vector. For higher dimensions, the norm is specified as the "size" (hard to visualise length in higher dimensions)

Theorem 1.1.2

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space. Then we have:

- $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ (Cauchy-Schwarz inequality)
- $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (Triangle inequality)

Proof: • If either $\mathbf{u}$ or $\mathbf{v}$ is $\mathbf{0}$, then $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\|\mathbf{u}\| \|\mathbf{v}\|$ are zero. So suppose $\mathbf{u}$ and $\mathbf{v}$ are non-zero. Let $\mathbf{w} = \mathbf{u} - \lambda \mathbf{v}$ where λ is a scalar. We know that $\langle \mathbf{w}, \mathbf{w} \rangle \geq 0$ so

$$\begin{aligned}
 0 &\leq \langle \mathbf{u} - \lambda \mathbf{v}, \mathbf{u} - \lambda \mathbf{v} \rangle \\
 &= \langle \mathbf{u}, \mathbf{u} \rangle - \lambda \langle \mathbf{v}, \mathbf{u} \rangle - \bar{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle + |\lambda|^2 \langle \mathbf{v}, \mathbf{v} \rangle \\
 &= \langle \mathbf{u}, \mathbf{u} \rangle - \lambda \langle \mathbf{v}, \mathbf{u} \rangle - \overline{\lambda \langle \mathbf{u}, \mathbf{v} \rangle} + |\lambda|^2 \langle \mathbf{v}, \mathbf{v} \rangle \\
 &= \langle \mathbf{u}, \mathbf{u} \rangle - 2\operatorname{Re}(\lambda \langle \mathbf{v}, \mathbf{u} \rangle) + |\lambda|^2 \langle \mathbf{v}, \mathbf{v} \rangle \\
 &= \|\mathbf{u}\|^2 - 2\operatorname{Re}(\lambda \langle \mathbf{v}, \mathbf{u} \rangle) + |\lambda|^2 \|\mathbf{v}\|^2
 \end{aligned}$$

Now let $\lambda = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}$. Then

$$\begin{aligned} 0 &\leq \|\mathbf{u}\|^2 - 2\operatorname{Re}\left(\frac{|\langle \mathbf{v}, \mathbf{u} \rangle|^2}{\|\mathbf{v}\|^2}\right) + \frac{|\langle \mathbf{v}, \mathbf{u} \rangle|^2}{\|\mathbf{v}\|^4} \|\mathbf{v}\|^2 \\ &= \|\mathbf{u}\|^2 - \frac{|\langle \mathbf{v}, \mathbf{u} \rangle|^2}{\|\mathbf{v}\|^2} \end{aligned}$$

Thus we have

$$0 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - |\langle \mathbf{v}, \mathbf{u} \rangle|^2 \iff |\langle \mathbf{u}, \mathbf{v} \rangle|^2 = |\langle \mathbf{v}, \mathbf{u} \rangle|^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2$$

Taking positive square roots gives the required answer

Note:-

Root function is increasing and one-to-one so this is true (not true that $x \leq x^2$, but true that $x^2 \leq y^2 \implies |x| \leq |y|$)

- By the Cauchy-Schwartz inequality,

$$\langle \mathbf{u}, \mathbf{v} \rangle \leq |\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Thus

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + 2\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| + \|\mathbf{v}\|^2 \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2 \end{aligned}$$

Thus $\|\mathbf{u} + \mathbf{v}\|^2 \leq (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$ and taking positive square roots gives us the desired result

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1.2 Orthogonality and Orthonormal Bases

Note:-

We turn now to something you are already familiar with, that is, the idea of an angle between vectors. In fact, the idea of an angle between vectors is not as important in general vector spaces as it is in $\mathbb{R}^2$ or $\mathbb{R}^3$. What does remain of great importance though, is the notion of perpendicularity or orthogonality. Recall from your first year course that two vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ are perpendicular if their dot product is zero. We extend this idea to any inner product space.

Definition 1.2.1

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V . We say $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ (their inner product is 0). We say $\mathbf{u} \perp \mathbf{v} \iff \langle \mathbf{u}, \mathbf{v} \rangle = 0$

Example 1.2.1

Example 5.2.2 Let P_2 have the inner product

$$\langle p, q \rangle = \int_{-1}^1 p(t)q(t) dt$$

and let $p(t) = t$ and $q(t) = t^2$. Then

$$\langle p, q \rangle = \int_{-1}^1 tt^2 dt = \int_{-1}^1 t^3 dt = 0$$

so the two polynomials are orthogonal. We also have

$$\|p\| = \sqrt{\langle p, p \rangle} = \sqrt{\int_{-1}^1 t^2 dt} = \sqrt{\frac{2}{3}}$$

and

$$\|q\| = \sqrt{\langle q, q \rangle} = \sqrt{\int_{-1}^1 t^4 dt} = \sqrt{\frac{2}{5}}.$$

It is easy to prove an analogue of the theorem of Pythagoras in an inner product space. Here it is:

Theorem 1.2.1

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V such that $\mathbf{u} \perp \mathbf{v}$. Then

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = \|\mathbf{u} + \mathbf{v}\|^2$$

: We have

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + 2\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \end{aligned}$$

⊕

Definition 1.2.2

Let S be a set of distinct vectors in an inner product space V

- S is said to be orthogonal if every pair of distinct vectors in the set S are orthogonal
- A vector $\mathbf{v}$ is said to be normal if $\|\mathbf{v}\| = 1$
- S is orthonormal if S is orthogonal and each member of S is normal

Note:-

The standard basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ of $\mathbb{R}^3$ (and more generally the standard basis for $\mathbb{R}^n$) is particularly convenient to work with. This can be attributed mainly to the fact that this basis forms an orthonormal set. Conversely, we'll see that in general orthogonal (and orthonormal) sets of vectors have many of the desirable properties of the standard basis in $\mathbb{R}^n$

Theorem 1.2.2

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are non-zero orthogonal vectors, then they are linearly independent

: Suppose $\lambda_1, \lambda_2, \dots, \lambda_n$ are scalars such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}$
We wish to show each $\lambda_i = 0$. Pick any $\mathbf{v}_i$, then

$$\begin{aligned} 0 = \langle \mathbf{0}, \mathbf{v}_i \rangle &= \langle (\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n), \mathbf{v}_i \rangle \\ &= \lambda_1 \langle \mathbf{v}_1, \mathbf{v}_i \rangle + \lambda_2 \langle \mathbf{v}_2, \mathbf{v}_i \rangle + \dots + \lambda_n \langle \mathbf{v}_n, \mathbf{v}_i \rangle \\ &= \lambda_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle \quad (\langle \mathbf{v}_j, \mathbf{v}_i \rangle = 0 \quad i \neq j) \end{aligned}$$

but $\langle \mathbf{v}_i, \mathbf{v}_i \rangle \neq 0 \implies \lambda_i = 0$

⊕

Definition 1.2.3

A basis B for a vector space V is orthogonal if the set B is orthogonal; it is orthonormal if B is orthonormal

Example 1.2.2

Example 5.2.7 The standard basis for $\mathbb{R}^n$ is an orthonormal basis.

The set of vectors

$$S = \{(1, 1, 1, 1), (-1, -1, 1, 1), (-1, 1, -1, 1), (-1, 1, 1, -1)\}$$

in $\mathbb{R}^4$ is orthogonal (check it yourself) but is not orthonormal since each vector in S has norm 2. By the last theorem, this set is linearly independent. It is therefore an orthogonal basis for $\mathbb{R}^4$. We could modify S to make it an orthonormal basis for $\mathbb{R}^4$ by dividing each vector by its norm. Thus

$$B = \left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2}(-1, -1, 1, 1), \frac{1}{2}(-1, 1, -1, 1), \frac{1}{2}(-1, 1, 1, -1) \right\}$$

is an orthonormal basis for $\mathbb{R}^4$.

We look now at an algorithm for transforming any (finite) basis of an inner product space into an orthonormal basis. The fact that we can do this means that any finite-dimensional inner product space has an orthonormal basis. The procedure we use is known as the Gram-Schmidt algorithm.

Theorem 1.2.3

Every (non-zero) finite-dimensional inner product space V has an orthonormal basis

: We do not prove this

⊕

The algorithm follows the following steps:

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be any basis for V . We give a rule for determining a set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ ($\mathbf{u}$ for unit) of orthonormal vectors (which must then be an orthonormal basis)

STEP 1: $\mathbf{v}_1 \neq \mathbf{0}$ so we set $\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$. Then $\mathbf{u}_1$ is normal

Recall the projection of $\mathbf{v}_2$ onto $\mathbf{u}_1$ is

$$\frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 = \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{u}_1$$

since $\|\mathbf{u}_1\| = 1$, then

$$\mathbf{w}_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{u}_1$$

is orthogonal to $\mathbf{u}_1$. Also $\mathbf{w}_2$ is non-zero, since if it were zero, we would have a non-trivial linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$ yielding the zero vector, which contradicts the linear independence

We set $\mathbf{u}_2 = \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|}$

STEP 3: The projections of $\mathbf{v}_3$ onto $\mathbf{u}_1$ and $\mathbf{u}_2$ are respectively

$$\langle \mathbf{v}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 \text{ and } \langle \mathbf{v}_3, \mathbf{u}_2 \rangle \mathbf{u}_2$$

If we subtract these two projections from $\mathbf{v}_3$ the resulting vector, $\mathbf{w}_3$, will be orthogonal to both $\mathbf{u}_1$ and $\mathbf{u}_2$, but only because $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthonormal. Let's see why:

$$\begin{aligned} \langle \mathbf{w}_3, \mathbf{u}_1 \rangle &= \langle \mathbf{v}_3 - \langle \mathbf{v}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{v}_3, \mathbf{u}_2 \rangle \mathbf{u}_2, \mathbf{u}_1 \rangle \\ &= \langle \mathbf{v}_3, \mathbf{u}_1 \rangle - \langle \mathbf{v}_3, \mathbf{u}_1 \rangle \langle \mathbf{u}_1, \mathbf{u}_1 \rangle - \langle \mathbf{v}_3, \mathbf{u}_2 \rangle \langle \mathbf{u}_2, \mathbf{u}_1 \rangle \\ &= 0 \end{aligned}$$

The check for $\mathbf{u}_2$ is similar. Also $\mathbf{w}_3$ cannot be the zero vector because of the linear independence of $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$. We set

$$\mathbf{u}_3 = \frac{\mathbf{w}_3}{\|\mathbf{w}_3\|}$$

STEPS 4 to n : Repeat as above with the obvious modifications

The result is a set of n orthonormal vectors, which must be a basis for the original space

Corollary 1.2.1

With the notation above, for each $k, 1 \leq k \leq n$, the set $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ spans the same subspace as the set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$

Example 1.2.3

Example 5.2.10 We apply the Gram-Schmidt procedure to the set of vectors

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \text{ in } \mathbf{R}^3, \text{ where } \mathbf{v}_1 = (1, 1, 1, 1), \mathbf{v}_2 = (0, 1, 1, 1), \mathbf{v}_3 = (0, 0, 1, 1)\}.$$

Notice first that S is a linearly independent set in $\mathbf{R}^4$. It must be a basis for $V = \text{span}(S)$, but $V \neq \mathbf{R}^4$ (why?). We find an orthonormal basis for V .

Since $\|\mathbf{v}_1\| = 2 \neq 1$, $\mathbf{v}_1$ is not normal. Put $\mathbf{u}_1 = \frac{1}{2}(1, 1, 1, 1) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Then $\|\mathbf{u}_1\| = 1$.

Put

$$\begin{aligned} \mathbf{w}_2 &= (0, 1, 1, 1) - \left\langle \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right), (0, 1, 1, 1) \right\rangle \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) \\ &= (0, 1, 1, 1) - \left(\frac{3}{4}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4} \right) = \left(-\frac{3}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right) \end{aligned}$$

Now $\|\mathbf{w}_2\| = \frac{1}{2}\sqrt{3}$, so

$$\mathbf{u}_2 = \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|} = \frac{\sqrt{3}}{6}(-3, 1, 1, 1).$$

For $\mathbf{u}_3$, first compute $\langle (0, 0, 1, 1), \mathbf{u}_1 \rangle$ and $\langle (0, 0, 1, 1), \mathbf{u}_2 \rangle$. They are, respectively, 1 and $\frac{1}{3}\sqrt{3}$. Then

$$\begin{aligned} \mathbf{w}_3 &= \mathbf{v}_3 - \langle \mathbf{v}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{v}_3, \mathbf{u}_2 \rangle \mathbf{u}_2 \\ &= (0, 0, 1, 1) - 1 \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) - \left(\frac{1}{3}\sqrt{3} \right) \left(-\frac{1}{2}\sqrt{3}, \frac{1}{6}\sqrt{3}, \frac{1}{6}\sqrt{3}, \frac{1}{6}\sqrt{3} \right) \\ &= (0, 0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) - \left(-\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6} \right) \\ &= \left(0, -\frac{2}{3}, \frac{1}{3}, \frac{1}{3} \right) \end{aligned}$$

You can check that $\|\mathbf{w}_3\| = \frac{1}{3}\sqrt{6}$, so set

$$\mathbf{u}_3 = \frac{\mathbf{w}_3}{\|\mathbf{w}_3\|} = \frac{1}{\sqrt{6}}(0, -2, 1, 1).$$

The three orthonormalized vectors are $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{u}_3$.